

Complex basics

Pull out i first: $\sqrt{-a} = i\sqrt{a}$. $\sqrt{-2} \cdot \sqrt{-6} = i^2\sqrt{12} = -2\sqrt{3}$.

Trap: $\sqrt{-2}\sqrt{-6} = \sqrt{12}$ flips the sign.

Powers of i : divide by 4, use the remainder: $R0 = 1$, $R1 = i$, $R2 = -1$, $R3 = -i$.

Negative power: add 4s until positive. i^{-13} : $-13 + 16 = 3$, so $i^3 = -i$.

Standard form: split and reduce *each* part: $\frac{-8+4i\sqrt{3}}{12} = -\frac{2}{3} + \frac{\sqrt{3}}{3}i$.

Subtracting: the minus flips *both* parts: $-(-7+i) = +7-i$.

Dividing: multiply top and bottom by the conjugate. Bottom becomes $a^2 + b^2$.

$$\frac{3+4i}{2-i} \cdot \frac{2+i}{2+i} = \frac{2+11i}{5} = \frac{2}{5} + \frac{11}{5}i.$$

Trap: bottom is $4+1=5$, not $4-1$.

Answer form: always $a+bi$; write $0-8i$, $-1+0i$. A finished answer can't be "divided down" ($-2-2\sqrt{3}i$ is not $-1-\sqrt{3}i$).

Quadratics: quadratic formula, $\sqrt{-36} = 6i$, then reduce: $x = 2 \pm 3i$.

Sum + graph: add parts; graph both as arrows from 0, resultant is the parallelogram diagonal. $|a+bi| = \sqrt{a^2+b^2}$.

Trig (polar) form

$a+bi \rightarrow r \operatorname{cis} \theta$: $r = \sqrt{a^2+b^2}$, θ from $\tan^{-1} \frac{b}{a}$, **then fix the quadrant** from the signs of a, b .

QII/QIII: $180^\circ \pm \text{ref}$. QIV: $360^\circ - \text{ref}$.

$r \operatorname{cis} \theta \rightarrow a+bi$: $a = r \cos \theta$, $b = r \sin \theta$.

Theorems

Product multiply r 's, **add** angles

Quotient divide r 's, **subtract** angles

Power (De Moivre) $(r \operatorname{cis} \theta)^n = r^n \operatorname{cis}(n\theta)$

Roots $\sqrt[n]{r} \operatorname{cis} \frac{\theta+360^\circ k}{n}$, $k = 0..n-1$

Bring every angle back into $[0^\circ, 360^\circ)$: past 360° subtract full turns; negative add 360° . In radians subtract 2π , **not** π .

Roots: isolate x^n first ($x^4 + 16 = 0 \Rightarrow x^4 = -16 = 16 \operatorname{cis} 180^\circ$). n roots, spaced $\frac{360^\circ}{n}$. Stop at trig form if asked.

Julia and Mandelbrot

Julia: $z^2 - 1$, then square that and $-1, \dots$ **Mandelbrot:** $z, z^2 + z, (z^2 + z)^2 + z, \dots$ (add the *original* z).

Take $| |$ each step. Exceeds 2 $\Rightarrow$ NO. Repeats/cycles $\Rightarrow$ YES.

Trap: exactly 2 is not exceeding: take another step. $(1 \pm i)^2 = \pm 2i$, not 2.

Polar points

$(r, \theta) \rightarrow (x, y)$: $x = r \cos \theta$, $y = r \sin \theta$. **Keep r 's sign.**

Negative r : face θ , walk backward (opposite quadrant).

$(x, y) \rightarrow (r, \theta)$: $r = \sqrt{x^2 + y^2}$, then quadrant-fix θ .

Equivalent points: same r : $\theta \pm 360^\circ$. Flip r : $\theta \pm 180^\circ$.

$(5, 120^\circ) = (5, -240^\circ) = (-5, 300^\circ) = (-5, -60^\circ)$.

Converting equations

Swaps: $x = r \cos \theta$, $y = r \sin \theta$, $x^2 + y^2 = r^2$.

Rect $\rightarrow$ polar: substitute, factor out r , solve: $3x - 2y = 6 \Rightarrow r = \frac{6}{3 \cos \theta - 2 \sin \theta}$. $x^2 + y^2 = 8x \Rightarrow r = 8 \cos \theta$.

Polar $\rightarrow$ rect, circle ($r = 6 \cos \theta$): multiply by r , substitute, complete the square: $(x-3)^2 + y^2 = 9$.

Line ($r = \frac{4}{\cos \theta - \sin \theta}$): clear fraction, substitute: $x - y = 4$.

Parabola ($r = \frac{2}{1 - \sin \theta}$): clear fraction, $r = y + 2$, square: $x^2 + y^2 = (y+2)^2 \Rightarrow x^2 = 4(y+1)$.

Line through two polar points

1) Convert both to (x, y) (watch negative r). 2) Slope. 3) Point-slope $y - y_1 = m(x - x_1)$. 4) Standard form $Ax + By = C$. 5) Substitute, factor r : $r = \frac{C}{A \cos \theta + B \sin \theta}$.

Intersections

Set $r = r$, solve on $[0, 2\pi)$, plug back for r . Don't divide by a trig function.

Then check the pole: if each curve hits $r = 0$ (at any angle), the pole is an intersection too. Setting $r = r$ never finds it.

Graph types (word bank)

Circle $r = a$, $r = a \cos \theta$, $r = a \sin \theta$ (no constant)

Cardioid $r = a \pm a \cos \theta$ (numbers equal); touches pole

Limaçon $r = a \pm b \cos \theta$, $a \neq b$. $a < b$: inner loop

Rose $r = a \cos n\theta$, $n \geq 2$. Odd n : n petals; even: $2n$. Length $|a|$

Lemniscate $r^2 = a^2 \cos 2\theta$ or $a^2 \sin 2\theta$; figure 8

cos: symmetric over x -axis. sin: over y -axis. **Trap:** $r = 7 \cos \theta$ is a circle, not a rose.

Calculator part (about 3 min)

Polar mode. **Degrees:** θ 0 to 360 (Graphing Angle setting is separate). Radians: 0 to 2π . Blank/sliver = mode and window mismatch.

r^2 equations: enter $r = \sqrt{\quad}$. Use Trace for key r values. Write your setup.